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The Pharmacy Diabetes Care Program: assessment of a community pharmacy diabetes service model in Australia

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Abstract

Aim To assess the impact of a community pharmacy diabetes service model on patient outcomes in Type 2 diabetes.

Methods The study utilized a multisite, control vs. intervention, repeated-measures design within four states in Australia. Fifty-six community pharmacies, 28 intervention and 28 control, were randomly selected from a representative sample of urban and rural areas. Intervention pharmacies delivered a diabetes service to patients with Type 2 diabetes, which comprised an ongoing cycle of assessment, management and review, provided at regular intervals over 6 months in the pharmacy. These services included support for self monitoring of blood glucose, education, adherence support, and reminders of checks for diabetes complications. Control pharmacists assessed patients at 0 and 6 months and delivered no intervention.

Results A total of 289 subjects (149 intervention and 140 control) completed the study. For the intervention subjects, the mean blood glucose level decreased over the 6-month study from 9.4 to 8.5 mmol/l ($P < 0.01$). Furthermore, significantly greater improvements in glycaemic control were seen in the intervention group compared with the control: the mean reduction in HbA_{1c} in the intervention group was –0.97% (95% CI: –0.8, –1.14) compared with –0.27% (95% CI: –0.15, –0.39) in the control group. Improvements were also seen in blood pressure control and quality of life in the intervention group.

Conclusion A pharmacy diabetes service model resulted in significant improvements in clinical and humanistic outcomes. Thus, community pharmacists can contribute significantly to improving care and health outcomes for patients with Type 2 diabetes. Future research should focus on clarifying the most effective elements of the service model.

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Keywords community pharmacy, health-care delivery, management, Type 2 diabetes

Abbreviations BMI, body mass index; BP, blood pressure; GP, general practitioner; LGA, local government area; QCPP, Quality Care Pharmacy Program

Introduction

Strict control of blood glucose in Type 2 diabetes can reduce the risk and delay the onset of the complications of diabetes [1], and bring about improvements in overall quality of life [2].

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As the number of cases of diabetes and the costs of care increase, the Australian health-care system will not be able to afford intensive management of Type 2 diabetes delivered solely by general practitioners (GPs) [3]. Hence, there is a need to develop innovative models of diabetes care which can increase patient access and reduce costs. Pharmacists, as highly trained and accessible health-care professionals, are well placed to develop an expanded role in diabetes care. The literature provides examples of positive health outcomes when pharmacists provide diabetes management services in research situations in both the clinic [4–8] and community pharmacy settings [9–12].

Specific services that pharmacists have offered include the provision of diabetes self-management education [4,6–8,10–12]; adherence support to medication [9,11]; ensuring the quality and evidence-based use of medications [4–8,10–12]; monitoring clinical outcome measures such as blood glucose levels [4–6,8–12], blood pressure [4,6,8,10,12], lipid levels [4–6,8–12]; and reminding patients of the importance of regular examinations for diabetic complications [10,11].

However, studies to date in community pharmacy settings have used either quasi-experimental designs [9–11] or a single pharmacist to deliver services within a randomized controlled trial [12], thereby limiting their generalizability. Thus, evidence of the effectiveness of pharmacy diabetes care models, derived from well-designed large-scale randomized controlled trials in community pharmacy, is still lacking. To address this gap, a study was designed to implement and assess a pharmacy diabetes service model in Australian community pharmacy.

Patients and methods

Approval for the study was given by the Human Ethics Committees of The University of Sydney (NSW), Monash University (Vic.), Curtin University (WA) and The University of Tasmania (Tas.). Written informed consent was obtained from all participating pharmacists and subjects. The study commenced in March 2004 and was completed in September 2004.

Study design

The study utilized a multisite, control vs. intervention, repeated-measures design. To detect a 0.5% reduction in HbA_{1c} in the intervention group compared with the control group post-intervention (SD 1.3%) with a power of 90% at the 5% significance level and allowing for a 20% dropout rate, at least 180 patients were required in each group [13].

Study sample

The sampling frame comprised all pharmacies accredited through the Quality Care Pharmacy Program (QCPP) [14], (a continuous quality improvement programme for community pharmacies in Australia), located within 300 km of the participating universities. The percentage of pharmacies in each local government area (LGA) was then calculated. To obtain a random stratified sample of 60 pharmacies, the corresponding percentage per LGA was calculated and the required number within each strata were randomly chosen using Excel™. Fifty-six pharmacies, including both urban and rural pharmacies, agreed to participate and were randomly allocated to the intervention or control group (28 control and 28 intervention).

All intervention pharmacists received a diabetes education manual for self-directed learning [15] and also attended a 2-day workshop. The workshop comprised a mixture of lectures on diabetes, pharmacotherapy, dietary management, and role-playing exercises; training on the use of the Medisense Optium™ meter and the Precision Link Direct Device™ (Abbot Diagnostics, North Ryde, Australia); insulin injection technique and devices; and blood pressure measurement using an Omron T5™ Digital Blood Pressure Monitor (Omron Electronics, North Ryde, Australia). Intervention pharmacists received reimbursement of $A200 (EUR 120) for each patient who completed all pharmacy visits over the study period.

Control pharmacists received training on recruiting patients, delivering the study protocol and documentation. They received a payment of $A20 (EUR 12) per patient who completed both baseline and final visits to the pharmacy for data collection.

Study subjects

Each pharmacy was asked to recruit 10 patients. The inclusion criteria to enter the programme were patients with established Type 2 diabetes and

- $HbA_{1c} \geq 7.5\%$, who were taking at least one oral glucose-lowering medication or insulin;
- $HbA_{1c} \geq 7.0\%$, who were taking at least one oral glucose-lowering medication or insulin *and* who were on at least one anti-hypertensive, angina or lipid-lowering drug.

Patients were recruited by the pharmacy using standardized brochures and signage, or through similar advertisments in local papers. Eligibility was verified by requesting the most recent clinical data (from within the previous 3 months) for HbA_{1c}, blood pressure (BP) and lipids from the patient's GP using a standardized form. These clinical data were requested again from the GP at the end of the study.

Study procedures

The service had a defined protocol and documentation procedure. Intervention pharmacists were provided with a patient file, which included the protocol for assessment, intervention and follow-up of each patient, worksheets for documentation of pharmacists' interventions and study questionnaires. The elements of the service included review of self monitoring of blood glucose; disease, medication, and lifestyle education; adherence support and detection of drug-related problems; and referrals to the patients' GPs when appropriate, e.g. for dosage or medication regimen adjustment when glycaemic and/or blood pressure control were suboptimal, and for checks for complications.

Intervention patients met with the pharmacist five times over 6 months. On the first visit with the pharmacist, the intervention patients were given a Medisense Optium™ blood glucose meter (Abbot Diagnostics, North Ryde, Australia), instructed in its use, and then asked to take measurements at least once daily (preferably at different times) for the duration of the study. If they already used a meter, they were asked to use the Medisense Optium™ meter instead for the duration of the study. At each subsequent visit, the patients' blood glucose measurements were downloaded onto the pharmacists' computers using the Precision Link Direct Device™ (Abbot Diagnostics, North Ryde, Australia) and the patient received a pie-chart printout which depicted their glycaemic control over the preceding period. The pharmacist used this as the basis for a discussion on the patients' progress and to identify other interventions to support patient care. These could include adherence support, medication review, and diabetes self-management and lifestyle information focusing on increasing physical activity and weight loss using approved national resources [16,17]. Individual goals were set at each visit, documented on a worksheet and reviewed at each subsequent visit. Patients were also referred to their physician as necessary.

The control patients had two visits with the pharmacist, one at the beginning and one at the end of the study. During the intervening 6 months, they received 'usual care' (i.e. no specialized diabetes service in the pharmacy).

Adherence to the study protocol was monitored and facilitated during visits by project staff to the pharmacies. In addition, pharmacists were contacted regularly by telephone and sent monthly newsletters to keep them informed and motivated. Separate newsletters were provided to control and intervention pharmacies.

Assessment of the service

The main outcome measure was change in HbA_{1c} as a marker of glycaemic control. Other clinical and humanistic parameters were also used to assess the service and to test the null hypotheses that there would be no significant difference between intervention and control groups pre- and post-intervention in:

- mean HbA_{1c}, systolic and diastolic blood pressure, lipid profile and body mass index (BMI);
- mean EQ-5D [18] quality-of-life scores.

Statistical analysis

All data were analysed using SPSS 10.0™ for Windows™ (SPSS, Chicago, Illinois, USA). Data are summarized as proportions, means ± SD, median (interquartile range) and mean differences (95% CI). Intervention and control groups were compared at baseline using an independent-samples Student's *t*-test or Mann–Whitney *U*-test for continuous parameters, and Pearson χ^2 test (with Yates' continuity correction in the case of dichotomous variables) for categorical parameters. Outcomes were analysed using repeated-measures multivariate ANOVA for continuous parametric parameters. For key variables with differences at baseline, a general linear model univariate ANCOVA was conducted using change in outcome as the dependent variable. For non-parametric continuous parameters, a Mann–Whitney *U*-test was used to compare changes between the control and intervention groups at 6 months. A two-tailed significance level of 0.05 was used in all analyses.

Results

A comparison of the demographic characteristics of both groups of pharmacies (number of staff, location) and pharmacists (age, gender and position in the pharmacy) confirmed their similarity and the success of the pharmacy randomization.

Three hundred and thirty-five eligible participants were initially recruited with 84% (149/176) of intervention patients and 88% (140/159) of control patients completing the study (Fig. 1). The most common reasons for non-completion included severe illness and moving to another location. The analysis of results used all available data for participants who fully completed the study and for non-completers for whom final clinical data were available (299 in total; 157 intervention, 142 control). Regrettably, it was not possible to retrieve final clinical data from GPs for all subjects, including some completers and some non-completers (72 in total: 33 intervention and 39 control). A comparison of completers with non-completers with respect to demographic, diabetes history and clinical parameters, including baseline HbA_{1c} in control subjects, showed no differences at baseline. In the intervention group, however, non-completers were younger than the completers (51 vs. 61 years; $P < 0.001$), had a higher mean baseline HbA_{1c} (9.6 vs. 8.8%; $P = 0.01$) and a higher mean BMI (35.2 vs. 31.8 kg/m^2; $P = 0.02$). No differences were found in other demographic or clinical parameters.

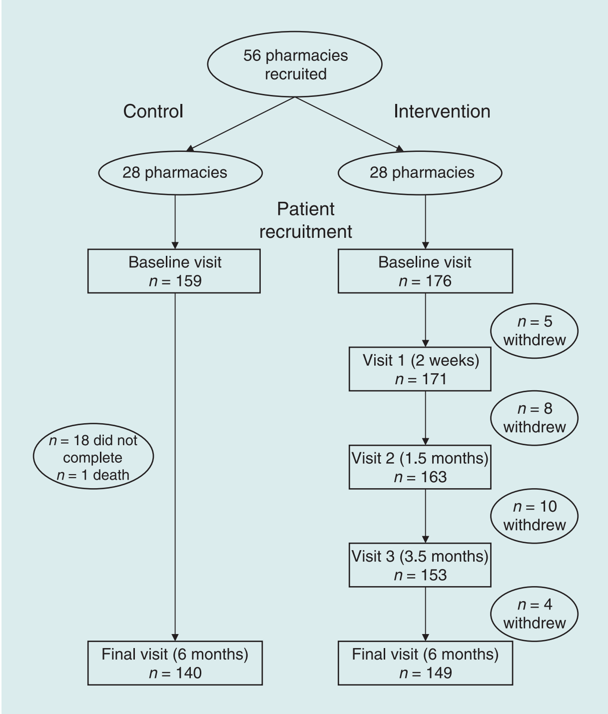

FIGURE 1 CONSORT diagram of DMAS recruitment and completion.

Study sample

Intervention and control patients were very similar in terms of demographics. The mean age was 62 ± 11 years and around half were male (51%). Overall, the medical histories of the two groups were also similar. The most common self-reported diabetes co-morbidities were hypertension (70%) and high cholesterol (61%). Approximately one-fifth of patients reported macrovascular and microvascular complications, such as angina and eye problems. Participants were taking a similar number of medications at baseline (6.8 ± 3.3 for the control subjects compared with 6.5 ± 3.0 for the intervention group).

Table 1 Clinical and humanistic parameters of participants at baseline and completion of study

		n	Baseline Mean ± SD	Final Mean ± SD	Mean difference (95% CI)	Intervention vs. control P-value*
HbA_{1c} (%)	Intervention	125	8.9 ± 1.4	7.9 ± 1.2	−1.0 (−0.8 to −1.3)	< 0.01
	Control	107	8.3 ± 1.3	8.0 ± 1.2	−0.3 (−0.003 to −0.5)	
BMI (kg/m^2)	Intervention	136	31.4 ± 5.9	31.1 ± 5.6	−0.4 (−0.8 to −0.01)	0.37
	Control	131	31.3 ± 6.7	31.1 ± 6.6	+0.2 (−0.1 to +1.3)	
Systolic BP (mmHg)	Intervention	87	135 ± 14	133 ± 15	−2.2 (−5.4 to +1.0)	0.06
	Control	92	133 ± 12	135 ± 15	+2.6 (−0.9 to +6.1)	
Diastolic BP (mmHg)	Intervention	87	79 ± 8	77 ± 8	−2.4 (−4.8 to −0.1)	0.52
	Control	92	77 ± 9	76 ± 9	−1.3 (−3.7 to +1.1)	
Total cholesterol (mmol/l)	Intervention	112	4.9 ± 1.1	4.7 ± 1.1	−2.1 (−4.1 to −0.02)	0.85
	Control	98	4.9 ± 1.0	4.7 ± 1.0	−2.1 (−4.4 to −0.01)	
Triglycerides (mmol/l)	Intervention	112	2.0 (1.4–2.8)	1.8 (1.3–2.5)	−0.3 (−0.05 to −0.5)	0.39†
	Control	97	1.8 (1.3–2.5)	1.7 (1.3–2.4)	−0.1 (−0.08 to −0.3)	
EQ-5D‡						
Utility score (range −0.6 to 1.0)	Intervention	143	0.8 (0.7–1.0)	0.8 (0.7–1.0)	−0.04 (−0.08 to +0.005)	0.07†
	Control	137	0.8 (0.7–1.0)	0.8 (0.7–1.0)	−0.02 (−0.04 to +0.03)	
Health-state scale (range 1–100)	Intervention	142	66.3 ± 19.5	71.6 ± 17.2	+5.3 (1.73 to 8.8)	0.02†
	Control	137	72.2 ± 18.9	73.3 ± 15.8	+1.1 (−1.6 to 3.8)	

Data are means ± SD, median (interquartile range). Changes over 6 months are mean difference (95% CI).
*Repeated measures multivariate ANOVA unless otherwise noted; †Mann–Whitney U-test on change scores; ‡quality-of-life scores [18].
BMI, body mass index; BP, blood pressure.

Similar proportions of intervention (92%) and control (88%) patients reported self monitoring of their blood glucose at baseline. Most patients (79%) reported being treated with oral glucose-lowering drugs alone; however, the proportion of patients taking a combination of insulin and oral glucose-lowering drugs was higher in the intervention than the control group (25 vs. 13%; $P = 0.01$). There was also a difference in years since diagnosis of diabetes, with the control group having been diagnosed with diabetes longer than the intervention group (10.4 vs. 8.6 years; $P = 0.04$). With respect to clinical measures, the control and intervention groups at baseline were similar with the exception of baseline HbA_{1c} ($P < 0.01$; Table 1).

Pharmacists' interventions

The service was delivered to patients during four visits to the pharmacy. Over the course of the four visits, pharmacists delivered a mean of 29 (SD ± 24) interventions per patient (an intervention refers to one type of information/education/assessment). Approximately one-third of these interventions related to SBGM (checking technique, strategies to address hypoglycaemia and hyperglycaemia); 31% to adherence to medications (education about medicines, ability to access medicines, and insulin administration); 29% to lifestyle (physical activity, nutrition, alcohol consumption and smoking) and foot-care issues; and 4% addressed checks of medication history (drug/dose discrepancies and potential therapeutic problems). Overall, 95% of intervention patients received interventions related to SBGM, 92% received lifestyle interventions and 89% received interventions related to adherence.

Clinical outcomes—intervention group

Blood glucose data were downloaded from the Optium™ meters of participants in the intervention group at each of the community pharmacy visits. Blood glucose data were not collected for control patients. Hence, all analyses in this section refer to comparisons within the intervention group. The mean blood glucose decreased significantly over the four visits from 9.4 mmol/l at the first visit to 8.5 mmol/l at the final visit ($P < 0.01$; Fig. 2a). In addition, the proportion of blood glucose readings which fell within the normal range significantly increased over the four intervention visits, from 39% at visit one to 51% at the final visit ($P < 0.01$; Fig. 2b). Systolic BP measured in the pharmacy also dropped from a mean of 143 mmHg at first visit to 137 mmHg at final visit ($P < 0.01$), and diastolic BP decreased from a mean of 82 mmHg at first visit to 79 mmHg at the final visit ($P = 0.05$).

Clinical outcomes for intervention and control groups

There was a significantly larger reduction in HbA_{1c} over the 6-month study in the intervention group (−0.97%: 95% CI: −0.8, −1.14) compared with the control group (−0.27%: 95% CI: −0.15, −0.39; Table 1). This difference between intervention and control groups was sustained when controlling for baseline differences in HbA_{1c}, years since diagnosis and type of pharmacotherapy (insulin vs. oral glucose-lowering agents; Table 2).

Systolic and diastolic BP measured by the GP decreased in the intervention group between baseline and final measurements,

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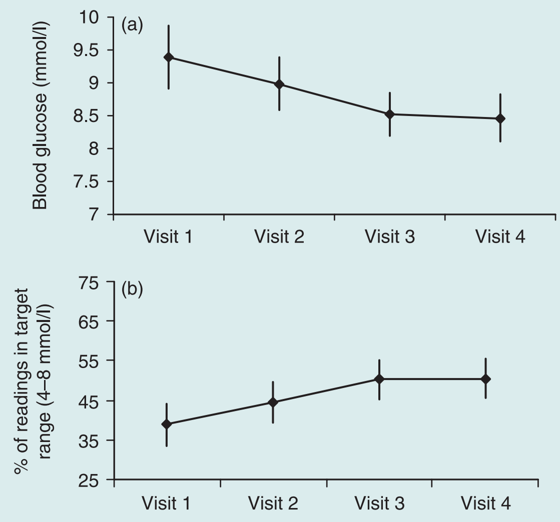

Figure 2 (a) Blood glucose readings (mean ± 95% CI) at the four pharmacy visits ($n = 123$, $P < 0.01$ comparing visit 1 to visit 4). (b) Percentage of blood glucose readings (mean ± 95% CI) within the target range (4–8 mmol/l) at the four pharmacy visits ($n = 123$, $P < 0.01$ comparing visit 1 to visit 4).

but when compared with the control group the decrease was not statistically significant (Tables 1 and 2). However, the percentage of patients who achieved target BP (130/80 mmHg) increased in the intervention group from 42% at baseline to 55% at final ($P = 0.07$), while the percentage of the control group decreased slightly (45 to 42%; $P = 1.00$). The BP measures reported in Table 1 were obtained from the participants' GP and differed from those made by the intervention pharmacists in the pharmacies.

Both intervention and control groups showed significant reductions in total cholesterol and triglycerides between baseline and the end of the study period (Table 1). The proportion of participants who exercised three or more times per week did not change significantly in either group (intervention group 54 to 62% ($P = 0.07$); control group 53 vs. 54%).

Humanistic outcomes

There were significant improvements in quality of life in the intervention group as indicated by increases in ED-5D health-state scale scores (Table 1). This improvement was sustained after controlling for baseline differences ($P = 0.02$; Table 2). The change in quality of life (EQ-5D utility score) was not significant ($P = 0.07$; Table 1).

Medication changes

The mean number of glucose-lowering medications taken increased from 1.8 at baseline to 2.0 in the intervention group with no change in the control group ($P = 0.04$).

Table 2 Multiple linear regression analysis of changes in main clinical outcomes after adjustment of key variables

	Regression coefficient for intervention group*	*P*-value
ΔHbA_{1c} (%)	−0.46	0.02
ΔSystolic BP (mmHg)	3.3	0.20
ΔDiastolic BP (mmHg)	−0.05	0.98
ΔBMI (kg/m^2)	0.65	0.12
ΔTotal cholesterol (mmol/l)	−0.11	0.43
ΔTriglycerides (mmol/l)	−0.04	0.83
ΔEQ-5D utility scores*	−0.03	0.36
ΔEQ-5D health-state scale scores*	0.39	0.03

Adjustment for diabetes duration, type of pharmacotherapy, baseline value [or ln(value)] for the associated variable.
*Quality-of-life scores [18].
BMI, body mass index; BP, blood pressure.

Discussion

A pharmacy diabetes service model for patients with Type 2 diabetes was successfully implemented in 28 community pharmacies in four States across Australia (New South Wales, Victoria, Western Australia and Tasmania). Implementation of the pharmacy diabetes service model resulted in improved health outcomes for patients including (i) better glycaemic control, based on significant improvements in mean blood glucose levels and reductions in HbA_{1c}; (ii) better BP control, based on reduction in mean systolic and diastolic BP readings; and (iii) improvements in quality of life. Strict control of diabetes can result in significant risk reduction in terms of the onset of complications [19,20]. Intensive blood glucose and BP control in patients with Type 2 diabetes have also been shown to be cost-effective in terms of managing these complications [21].

As the pharmacist delivered a multicomponent intervention, it is difficult to pinpoint the most effective elements that contributed to improved glycaemic control. However, it is likely that the combination of improved medication adherence, small increases in physical activity and modest weight loss all contributed to the results. It is also possible that the increased contact with the pharmacists contributed to the observed improvements.

It is interesting to note that there was a small, but statistically significant, reduction in HbA_{1c} in the control group. There were also significant reductions in total cholesterol and triglycerides in both the intervention and control patients. These improvements observed in the control patients might be as a result of the pharmacists' 'usual care' in combination with the recruitment process. The recruitment process involved asking patients to complete questionnaires and required clinical data from the GP, which may have prompted additional care from both GPs and pharmacists.

In excess of 400 patients expressed interest in taking part, but, of these, approximately 25% did not meet the eligibility criteria. Thus, a total of 335 eligible patients (176 intervention and 159 control) were fully recruited into the study and of these 84% of intervention and 88% of control patients completed the study. Overall, the intervention and control pharmacies and pharmacists were well matched in terms of pharmacy and personal demographics. Study participants were also well matched with respect to demographics, diabetes history and most clinical parameters.

The success of the model supports findings in similar community pharmacy-based, pharmacist-delivered diabetes programmes, where improvements in clinical outcomes have been reported [12,22,23]. In our study, we observed a large decrease of 0.97% (95% CI: –0.8, –1.14) from baseline to final HbA_{1c} in the patients who received the 6-month pharmacy diabetes service, which is very similar in magnitude to the findings of Cranor *et al.* [22] and Wermeille *et al.* [23]. However, it was double the improvement in HbA_{1c} reported by Clifford *et al.* [12] in a 12-month pharmacist-delivered pharmaceutical care programme for Type 2 diabetes, and double the improvement observed in our earlier research [11].

An important reason for this difference in effects between studies is undoubtedly as a result of differences in baseline HbA_{1c} of study participants. Choe *et al.* [24] reported that the response to a pharmacist-delivered clinic intervention varied according to baseline HbA_{1c}, i.e. the higher the baseline, the greater the percentage reduction.

Other reasons for differences in clinical impact between studies may be the nature and intensity of the pharmacist-delivered intervention. The service implemented in this study incorporated a complex service model, which included supporting SBGM, providing adherence support, providing advice about medication issues and instruction on lifestyle issues. Pharmacists utilized a range of interventions in the delivery of the service (average of 29 per patient) which were tailored to the needs of the individual patient. Ongoing monitoring of the patients' progress was seen as an essential element of the management process. The majority of trials of diabetes care models delivered by pharmacists, either in clinic or community settings, have utilized varied and complex interventions [4–12] and it is unclear to what extent individual elements of these contributed to the effectiveness of the service.

Study limitations

Certain limitations are associated with this study. Whilst the intervention and control groups were well matched on most clinical parameters, HbA_{1c} was higher at baseline in the intervention compared with the control group (8.9 vs. 8.3%). To address the uncertainty that may arise as a result of this, we used appropriate statistical methods to control for this baseline difference. There were also difficulties in retrieving final clinical data from GPs for some patients; however, the proportion of patients for whom final clinical data were missing was similar for intervention (20%) and control groups (24%). The duration of the study was only 6 months and further research is needed to clarify the sustainability of changes in disease control over time and the type and intensity of interventions that are most clinically and cost-effective.

Conclusion

This study is the first large-scale randomized controlled trial to determine the impact of a diabetes service model delivered by community pharmacists. As such, it represents a milestone in community pharmacy practice research and considerably strengthens the evidence base for the value of diabetes management services delivered in community pharmacy. The results of this study demonstrate that community pharmacists, as highly trained and accessible health-care professionals, can provide a clinically effective professional service for people with Type 2 diabetes.

Competing interests

None to declare.

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